

**Vishwakarma Government Engineering College, Chandkheda**  
**Department of Information & Communication Technology**  
**Semester- VI**  
**Analog and Digital Communication (3163206)**  
**Assignment 1**

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1. Define the following terms:
  - a) Communication system
  - b) Modulation
  - c) Noise figure
  - d) Modulation index (AM)
  - e) Frequency deviation
  - f) Pre-emphasis
2. List the elements of a basic communication system with a neat labeled diagram.
3. State the need for modulation in communication systems.
4. Write the expression of AM wave and define each term.
5. Explain different types of noise in communication systems with suitable examples.
6. Explain the frequency spectrum of sinusoidal FM and the significance of Bessel functions.
7. Discuss practical issues in FM transmission and the need for pre-emphasis and de-emphasis.
8. An AM transmitter radiates 10 kW power when modulation index is 0.8. Calculate:
  - a) Carrier power
  - b) Sideband power
  - c) Total sideband power
9. Calculate the noise figure of a system if input SNR is 40 dB and output SNR is 30 dB.
10. Compare AM, DSB-SC and SSB-SC in terms of:
  - a) Bandwidth
  - b) Power efficiency
  - c) Complexity
  - d) Noise performance
11. Analyze the effect of increasing modulation index in AM system. What happens if  $m > 1$ ?
12. Evaluate the advantages and limitations of SSB over conventional AM.
13. Explain the working of amplitude demodulator circuit with the remedy for avoiding diagonal peak clipping and negative peak clipping
14. What are coherent digital modulation techniques?
15. For the FM System, the modulating frequency is 4 KHz and the maximum frequency deviation is 20 KHz. Calculate modulation index  $\beta$  and bandwidth of the system using Carson's Rule.
16. The modulating signal in an FM wave is 500 Hz with amplitude of 3.2 Volt and frequency deviation is 6.4 kHz. If the audio freq. voltage is now increased to 8.4 volt determine the new frequency deviation and modulation index.
17. Draw the block diagram of Superheterodyne Receiver and explain its operation.

18. State Friss Formula for overall Noise Factor of amplifiers in cascade and Using Friss Formula, Find Overall Noise Figure of three stage cascaded amplifier, each stage having a power gain of 10 dB and noise figure of 5 dB.
  19. What do you mean by VSB Modulation and Why it is used?
  20. Discuss Noise factor. Derive expression for noise factor of amplifier connected in cascade
  21. Explain Armstrong method of FM generation with neat diagram.
  22. The antenna current of an AM transmitter is 8 A for an unmodulated carrier but it increases to 8.93 A when the carrier is modulated by a single sine wave. Find the percentage modulation. Determine the antenna current when the percentage modulation changes to 0.8.
  23. List the methods used to generate SSB modulation signal .discuss any one method in detail.
  24. Discuss the basic principal of FM detection and explain Foster-seeley discriminator.
  25. Draw the block diagram of Tuned radio Frequency (TRF) Receiver and explain its operation. Describe the problems in TRF receiver.
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